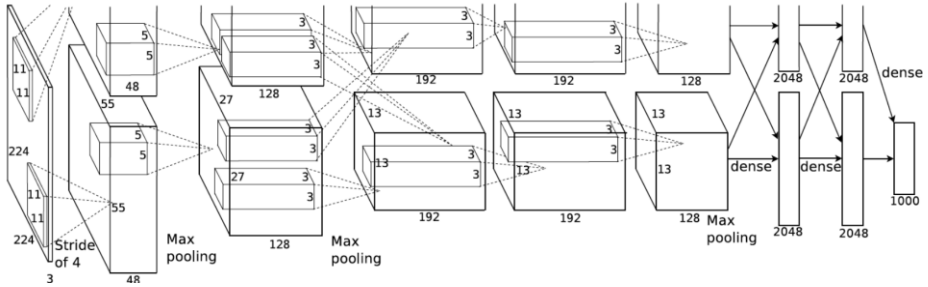
Overparametrized Modeling

2012

AlexNet

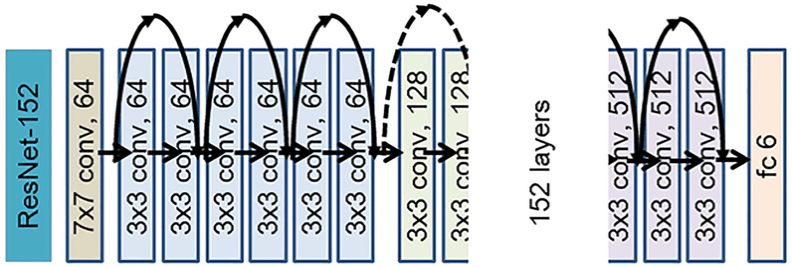


~60 million parameters

ImageNet **~1 million images**

2020

ResNet 152x4



~1 billion parameters

JFT dataset **300 million images**

[Krizhevsky et al., "ImageNet Classification with Deep Convolutional Neural Networks", NIPS 2012]
[Kolesnikov et al., "Big Transfer (BiT): General Visual Representation Learning", ECCV 2020]

Beyond Standard Labels

Images can be “cheaply” acquired

- Natural images can be found in abundance on the web
  - **Flickr** includes billions of images. In 2016, ~**2 million** images were uploaded to Flickr everyday
  - **300,000,000** photos uploaded to **Facebook** everyday. Total number of Facebook photos as of 2015: **250,000,000,000**
  - 300 hours of video are uploaded to **YouTube** every minute!
    - Assuming 1 useful frame per second, **everyday** there is an additional **1,555,200,000** images on youtube
  - In 2014, according to Mary Meeker's annual Internet Trends report, people uploaded an average of **1.8 billion** digital images every single day. That's 657 billion photos per year.
  - Simple estimations (e.g. InfoTrends) suggests that we have ~1.5 trillion images taken in 2017.
  - …
- Special images can usually be found in large amounts
  - Massive ditigitization in the past couple of decades in national organizations (hospitals, city office, etc.)
  - Affordable, small, and high-definition sensors that can be used by companies
  - …



Beyond Standard Labels

Annotation and Labelling is a Laborious Task

- **Expensive** (especially when require expert opinion, e.g., MD specialists)
- **Time-Consuming**
- **Noisy** (due to human error, ill-definition, limited annotation time, etc.)



Learn from massive datasets with low annotation effort



Content

- Self-supervised learning
  - Principle of understanding natural structures
  - Metric Learning
- Semi-supervised learning
  - Pretraining
  - Principle of Continuity
  - Principle of Consistency
- Knowledge transfer
  - Transfer Learning
  - Domain Adaptation
  - Knowledge Distillation
- Learning with noisy labels
  - Robustness to label noise
  - Identifying and removing noisy samples
  - Correcting label noise

